Function transformations

1 Translate the graph up by 4 units.

2 Translate the graph to the left by 6 units.

3 Move the graph to the left by 5 units.

4

a $y = 4x + 2$

b $y = (x - 10)^3$

c $y = 2^x - 7$

5

a $-9, 2$

b $-1, 10$

c $-5, 6$

d $5, -6$

6

a 4

b 4

c 3

d 5

7

a $f(x) - 1$

b $f(4x)$

c $3f(x)$

8

a $g(x) = f(x) - 3$

b $g(x) = \dfrac{3}{x} - 3$

9

a $g(x) = f(x + 4)$

b $g(x) = 3^{x+4}$

10

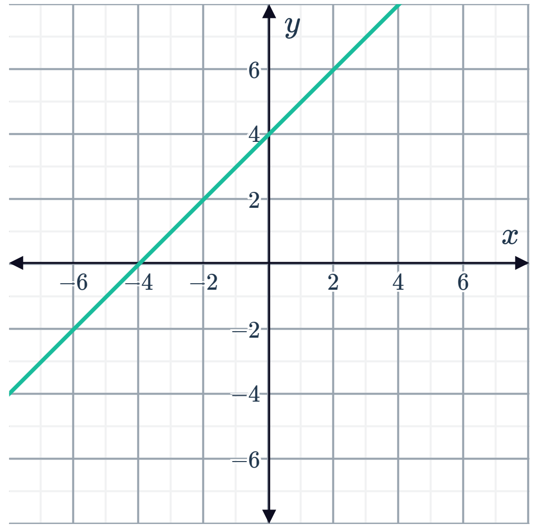
a $y = x^2 + 2$

b $y = (x - 3)^2 - 3$

11

a $g(x) = x + 4$

b



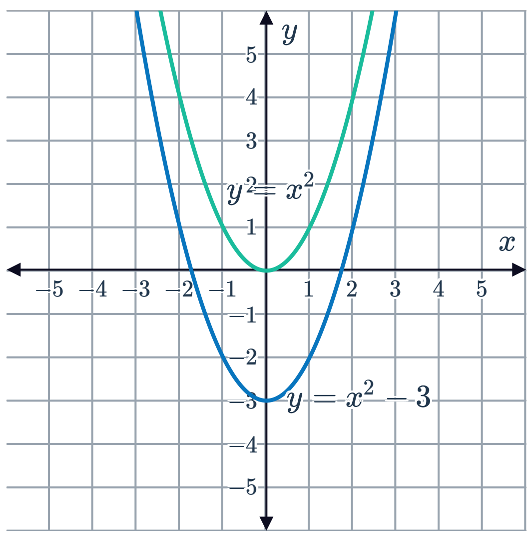
c A translation of 4 units up

12

a

i Move the graph downwards by 3 units.

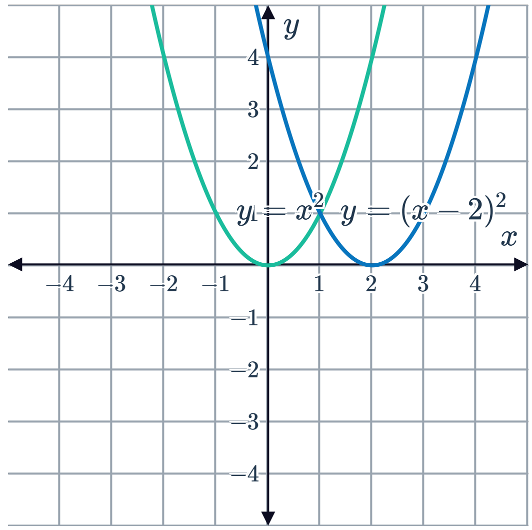
ii



b

i Move the graph to the right by 2 units.

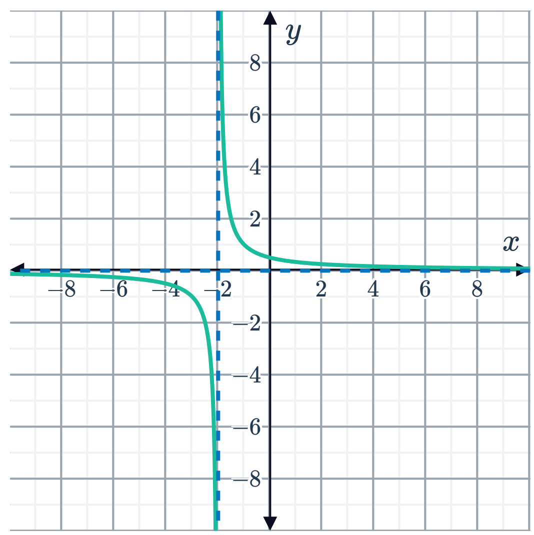
ii



13

a Shift the graph to the left by 2 units.

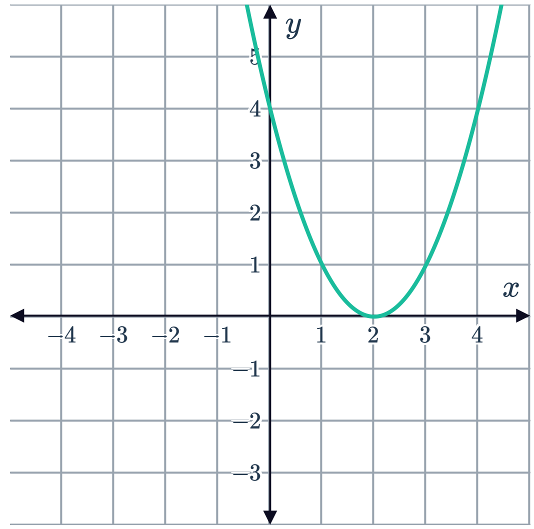
b



14

a Move the graph to the right by 2 units.

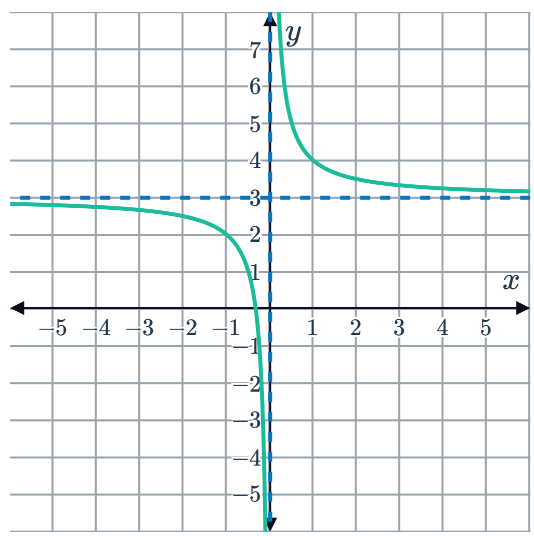
b



15

a Move the graph upwards by 3 units.

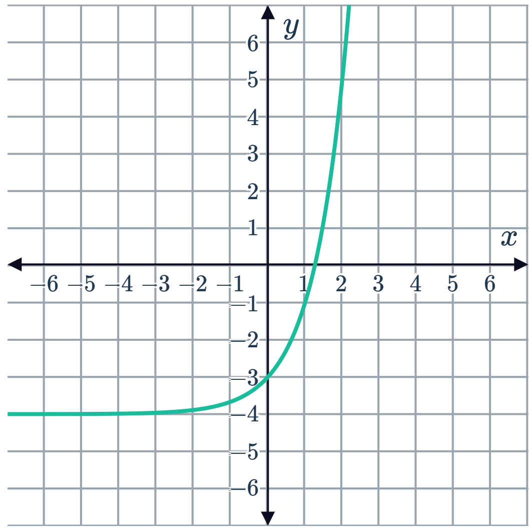
b



16

a Translate the graph downwards by 4 units.

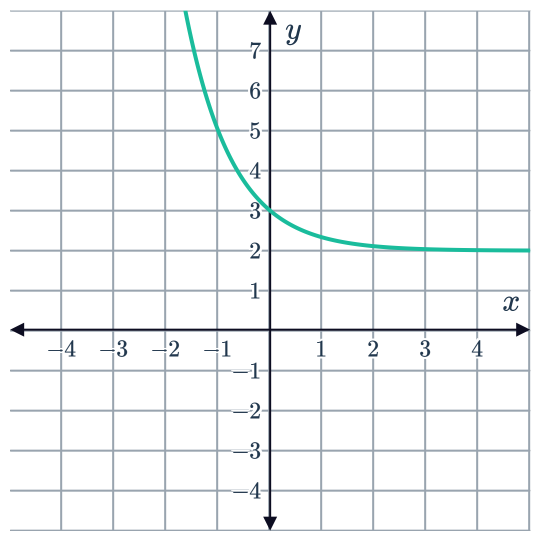
b



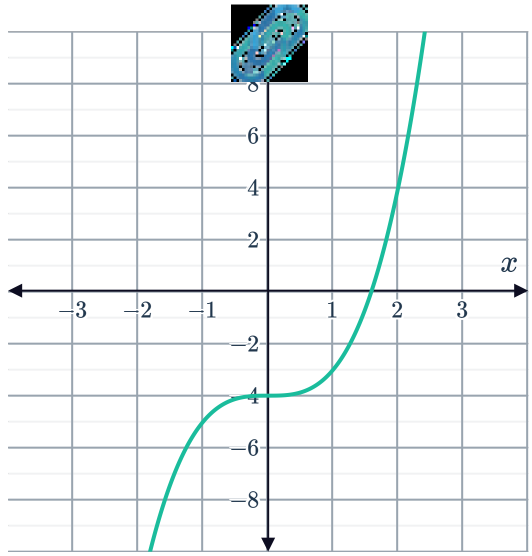
17

a Move the graph upwards by 2 units.

b



18



19

a $f(x) = x^2$

b $g(x) = \frac{1}{4} f(x)$

c $g(x) = \frac{1}{4} x^2$

d $h(x) = -f(x)$

e $h(x) = -x^2$

Multiple transformations

20

a $f(x)$ undergoes a vertical stretch by a factor of 5, is reflected across the horizontal axis and is translated 8 units up.

b $f(x)$ is translated 4 units left, undergoes a vertical stretch by a factor of 7 and is reflected across the horizontal axis.

c $f(x)$ is translated 8 units left, undergoes a vertical stretch by a factor of 10, is reflected across the horizontal axis and is translated 9 units down.

21 $y = -(x-6)^2 + 5$

22

a $(-11, -3)$

b $(0, -12)$

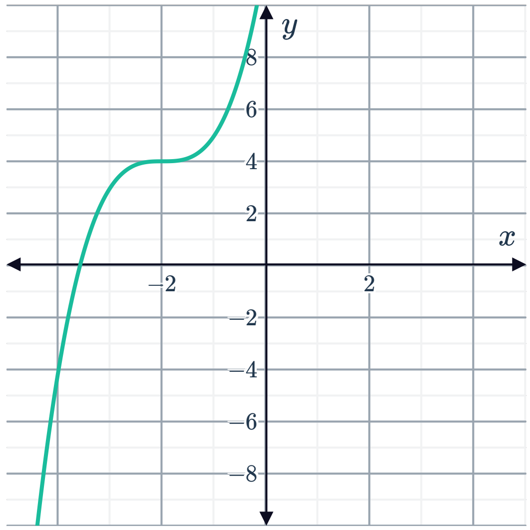
c $\left(-\frac{5}{6}, 3\right)$

23 $g(x) = \frac{1}{3}\sqrt{2x}$

24

a Move the graph to the left by 2 units and up by 4 units.

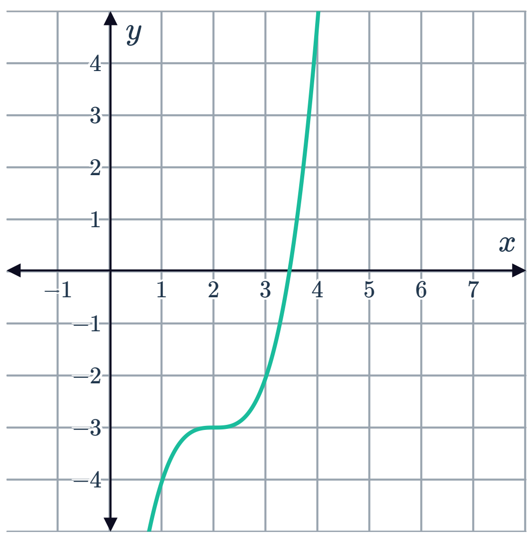
b



25

a Move the graph to the right by 2 units and down by 3 units.

b



26

a Starting with $y = x^2$, the function undergoes a vertical stretch by a factor of 3, is reflected across the horizontal axis and is translated 4 units down.

b $y = -3x^2 - 4$

27

a

i Starting with $y = x^3$, the function is translated 3 units right and undergoes a vertical stretch by a factor of 2.

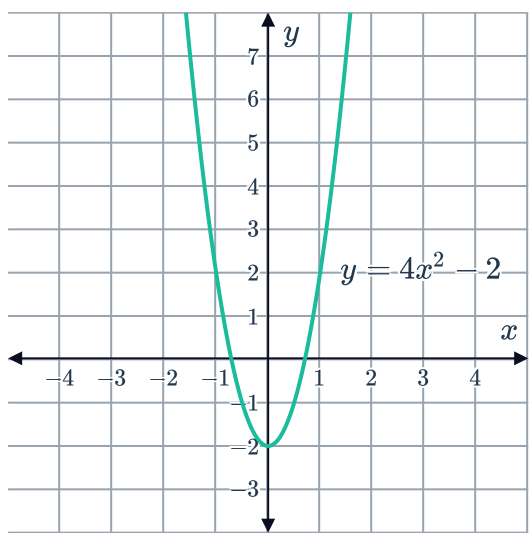
ii $y = 2\left(x - 3\right)^3$

b

i The function is translated 3 units left, undergoes a vertical stretch by a factor of 3, is reflected across the horizontal axis and is translated 2 units up.

ii $y = -3\left(x + 3\right)^3 + 2$

28

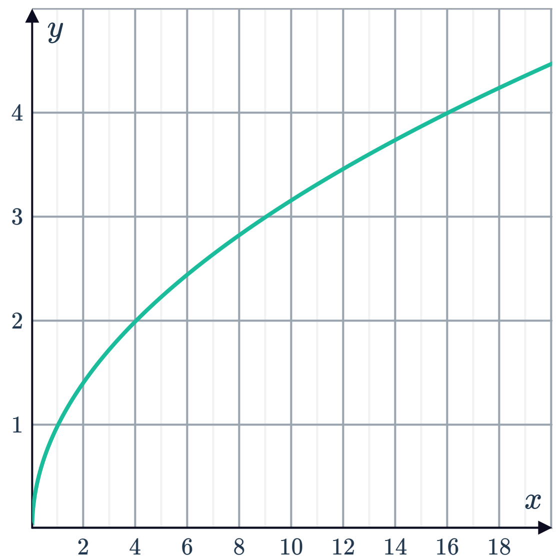


29

a

x	0	1	4	9	16
$f(x)$	0	1	2	3	4

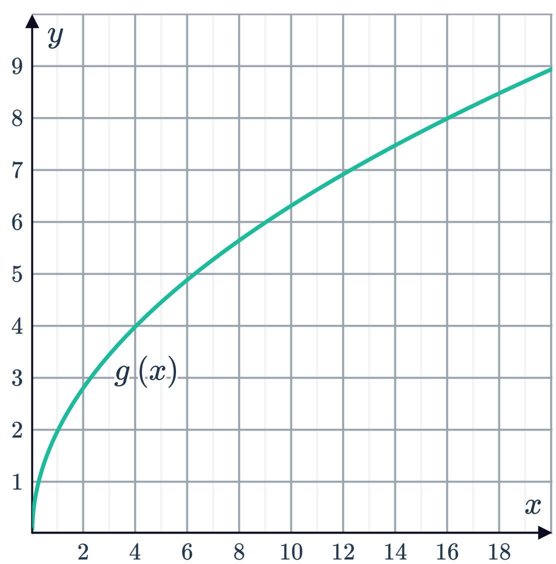
b



c

x	0	1	4	9	16
$f(x)$	0	1	2	3	4
$g(x)$	0	2	4	6	8

d



e The graph of $g(x)$ can be considered as the graph of $f(x)$ stretched horizontally by a factor of 4, or compressed vertically by a factor of $\frac{1}{4}$.

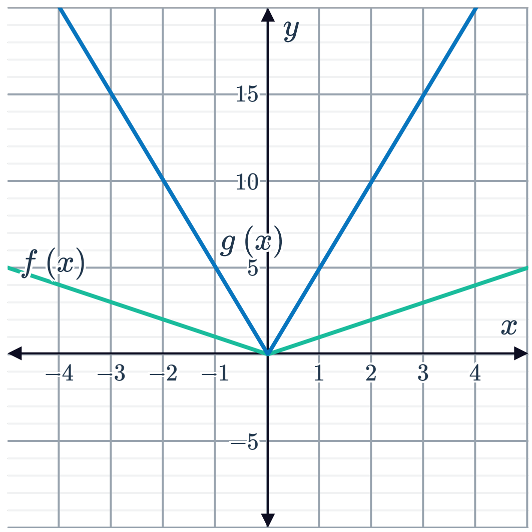
f The square root of negative numbers is undefined.

30

a

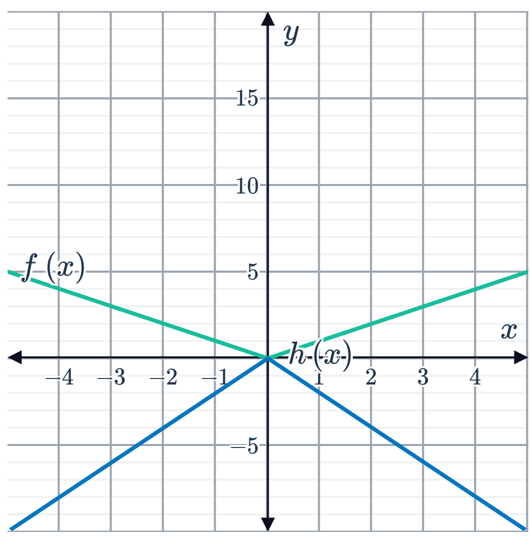
x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3
$g(x) = 5f(x)$	15	10	5	0	5	10	15
$h(x) = -2f(x)$	-6	-4	-2	0	-2	-4	-6

b



c Vertically stretched by a factor of 5.

d



e Vertically stretched by a factor of 2 and reflected across the x-axis.